

Optimal Immigration and Long-Run Growth

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Outline

1. What the paper does
2. The planner's problem and the extended Golden Rule
3. The positive-immigration condition
4. Two-period OLG extension and decentralization
5. Other considerations

The paper turns immigration policy into a Golden Rule growth problem:

- ▶ immigration is optimal when the marginal output and demographic benefits exceed immigrant consumption, public-good costs, and capital requirements for future natives

Benchmark framework

A benchmark

- ▶ Closed neoclassical growth economy
- ▶ Natives and immigrants have identical productivity
- ▶ Immigrants are available in unlimited supply
- ▶ Public goods are rival and non-excludable

Welfare criterion

- ▶ Maximize steady-state utility of a representative native
- ▶ Immigrants receive at least a threshold utility/consumption level
- ▶ Children born in the destination country are treated as natives

Not a global welfare criterion but a host-country Golden Rule benchmark.

Baseline demographics and resource constraint

Workers

$$L_t = N_t + M_t, \quad m_t \equiv \frac{M_t}{N_t}.$$

Each worker has $1 + \gamma$ children, and children born in the country are natives:

$$N_{t+1} = (1 + \gamma)(N_t + M_t).$$

Steady-state growth

$$\Gamma(m) = (1 + \gamma)(1 + m).$$

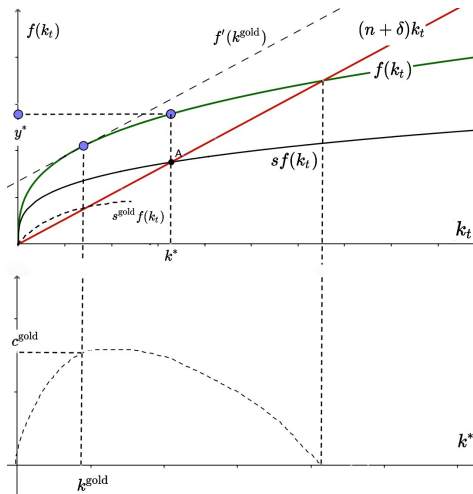
In steady state, natives, immigrants, labor, and capital all grow at $\Gamma(m)$.

Planner's choice

- Choose the capital-labor ratio k and immigrant-native worker ratio m to maximize native steady-state consumption/utility, subject to immigrant participation.

Golden Rule

The 'standard' Golden Rule equates the return on capital to the economy's growth rate. Solow:



Main result here: an extended Golden Rule

Here the growth rate is partly chosen through immigration, but same logic.

Extended Golden Rule

$$1 + f'(k) - \delta = \Gamma(m) = (1 + \gamma)(1 + m).$$

No immigration

$$m = 0 \quad \Rightarrow \quad f'(k_G) - \delta = \gamma.$$

This is the Phelpsian Golden Rule.

With immigration

$$m > 0 \quad \Rightarrow \quad \Gamma(m) > 1 + \gamma.$$

Higher chosen growth implies a lower Golden Rule capital-labor ratio.

Immigration endogenizes population growth, but the Golden Rule logic survives.

When is positive immigration optimal?

In the one-period model, each immigrant requires consumption ξ .

Define c_G and k_G as the no-immigration Golden Rule consumption and capital-labor ratio.

Sufficient condition for $m^* > 0$:

$$\underbrace{f(k_G) - k_G f'(k_G)}_{\text{marginal product of labor}} > \underbrace{\xi + g}_{\text{output to immigrant}} + \underbrace{(1 + \gamma)k_G}_{\text{new capital needed for immigrant children}}.$$

or

$$c_G > \xi + (1 + \gamma)k_G.$$

What would it take to compare to U.S. data?

The model's immigration variable is the immigrant-native worker ratio for one model period:

$$m_t = \frac{M_t}{N_t}, \quad \Gamma(m_t) = (1 + \gamma)(1 + m_t).$$

- ▶ Construct in the data: $\hat{m}_T \approx \sum_{s=1}^T \text{WA NIM}_s / N_0^W$. WA NIM is working-age net international migration; N_0^W is initial native workers.
- ▶ Choose T to match the model period.
- ▶ U.S. data could discipline $\hat{m}_T$, g , labor's share, k/y , ξ (?).

U.S. calibration targets

Object	Value	Range
Labor share, $1 - \alpha$	0.62	0.60–0.65
Depreciation rate, δ	0.10	0.04–0.10
Natural population growth, γ	0.0015	0–0.003
Government purchases, g/y	0.17	0.10–0.20
Foreign-born/native labor force	0.24	—
Approx. annual WA NIM/native workers	0.007–0.015	policy-dependent
Threshold, ξ/y (observed income)	0.52	0.35–0.70

Interpretation

Use observed immigrant labor income to scale ξ : BLS foreign-born earnings are about 86% of native-born earnings, so $\xi/y \approx 0.86 \times (1 - \alpha) \approx 0.53$; the check below uses 0.52.

Sources: BEA government consumption and investment; Census population estimates; BLS foreign-born labor force and earnings.

Back-of-the-envelope Golden Rule check

With Cobb-Douglas production and annual U.S. targets,

$$\frac{k_G}{y} = \frac{\alpha}{\delta + \gamma} \approx \frac{0.38}{0.1015} = 3.7, \quad \frac{c_G}{y} = 1 - \alpha - \frac{g}{y} \approx 0.45.$$

Interpret one model period as a 30-year working generation. Then the sufficient condition becomes

$$(30) \left(\frac{c_G}{y} - \frac{\xi}{y} \right) > (1 + \gamma)^{30} \frac{k_G}{y}.$$

This is an annualized sufficient-condition check, not a structural estimate of the full m^* .

Immigrant threshold	Surplus flow	Capital cost	Suff.?
$\xi/y = 0.35$	$30(0.45 - 0.35) = 3.0$	3.9	no
$\xi/y = 0.52$	$30(0.45 - 0.52) = -2.1$	3.9	no
$\xi/y = 0.70$	$30(0.45 - 0.70) = -7.5$	3.9	no
Cutoff: $\xi/y < 0.32$	> 3.9	3.9	yes

Under an observed-income interpretation of ξ , $\xi/y \approx 0.52$ is well above the cutoff. The sufficient condition would require an immigrant threshold below about 32% of native income.

Why closed-economy capital matters

The sufficient condition is not just about immigrant consumption or public goods. In a closed economy, the extra future native population also has to be equipped with capital.

$$c_G > \xi + \underbrace{(1 + \gamma)k_G}_{\text{capital for immigrants' children}}$$

- ▶ Without the capital term, the rough cutoff would be $\xi/y < c_G/y \approx 0.45$.
- ▶ With the 30-year capital term,

$$\frac{\xi}{y} < 0.45 - \frac{(1 + \gamma)^{30}}{30} \frac{k_G}{y} \approx 0.32.$$

- ▶ If capital were internationally mobile, or immigrants arrived with capital, this marginal cost would be much smaller.

The closed-economy capital requirement is central to why the sufficient condition is empirically demanding.

Does lower fertility raise optimal immigration?

In the sufficient condition, lower natural population growth γ slightly raises the cutoff value of ξ/y that would imply positive immigration:

$$\frac{\xi}{y} < \frac{c_G}{y} - \frac{(1+\gamma)^{30}}{30} \frac{k_G}{y}, \quad \frac{k_G}{y} = \frac{\alpha}{\delta + \gamma}.$$

Natural population growth, γ	sufficient cutoff for ξ/y
0.0030	0.315
0.0015	0.319
0.0000	0.323
-0.0050	0.335
-0.0100	0.346

- ▶ Lower fertility makes immigration more attractive at the margin.
- ▶ Empirically this is small relative to the observed-income calibration, $\xi/y \approx 0.52$.

Two-period OLG extension

The paper moves to a Diamond-style OLG setting: people work when young and consume when young and old.

- ▶ **Same core result:** $1 + f'(k) - \delta = \Gamma(m)$. The extended Golden Rule is unchanged.
- ▶ **New demographic channel:** dependency ratio = old/young = $1/\Gamma(m)$. More immigration raises the young-working population relative to retirees.

Why this matters

Immigration now affects both production and the intertemporal allocation of consumption across young and old natives.

What changes in the OLG version?

The extended Golden Rule is unchanged, but the marginal immigration condition is not.

Interior OLG condition:

$$\underbrace{f(k) - kf'(k)}_{MB(m)} = \underbrace{X(m) + g + \Gamma(m)k}_{\text{direct marginal cost}} - \underbrace{\frac{\beta}{1+m} \{C(m) + mX(m)\}}_{\text{dependency-ratio benefit}}.$$

- ▶ $X(m)$ is the immigrant consumption cost consistent with threshold utility.
- ▶ Higher m lowers the old-young dependency ratio and raises the utility value of a given consumption bundle.
- ▶ This is the main insight beyond the one-period Golden Rule benchmark.

Decentralization: implementation intuition

The planner allocation can be supported competitively by using fiscal policy to reproduce planner choices

- ▶ **Participation:** the entry transfer leaves immigrants exactly at their threshold utility, so they are willing to come but do not capture extra surplus.
- ▶ **Public-goods cost:** a uniform consumption tax finances g without changing the intertemporal price faced by young and old consumers.
- ▶ **Capital allocation:** government bonds absorb or supply saving so private portfolios deliver the planner's capital stock $k(m^{**})$.

At the extended Golden Rule, $1 + r = \Gamma(m^{**})$. Bonds implement the capital allocation; they do not finance steady-state government spending.

Discussion

- ▶ Clear benchmark for “optimal immigration”
- ▶ Makes the native-welfare criterion explicit
- ▶ Model makes clear what must be paid: immigrant cons., public goods, and capital for descendants
- ▶ Makes clear what can offset those costs: immigrant output (and a lower dep ratio in OLG)
- ▶ Connects immigration to the Golden Rule literature

Other points

- ▶ Native-only steady-state welfare is explicit (migrants to natives?)
- ▶ Closed-economy capital accumulation does much of the work in the result (discussed)
- ▶ The immigrant participation constraint is doing the redistributorial work
- ▶ No skill heterogeneity, complementarities, or endogenous assimilation
- ▶ *Transition dynamics* and cohort incidence are mostly outside the analysis

Participation constraint and surplus

Immigrants enter the planner's problem through a participation constraint, not through the native welfare objective:

$$\max_{k,m,x} C(k, m) - mx \quad \text{s.t.} \quad x \geq \xi.$$

- ▶ Since x is a cost to natives, the planner sets immigrant consumption at the threshold: $x = \xi$.
- ▶ Any surplus above the immigrant participation threshold is assigned to natives.
- ▶ This is why the marginal cost of immigration contains ξ , rather than immigrant marginal product or immigrant utility.

The clean native-welfare benchmark relies on an entry-transfer logic: immigrants are made just willing to come.

What the benchmark abstracts from

The model treats immigration mainly as changing labor quantity and population growth, not the composition or productivity structure of the economy.

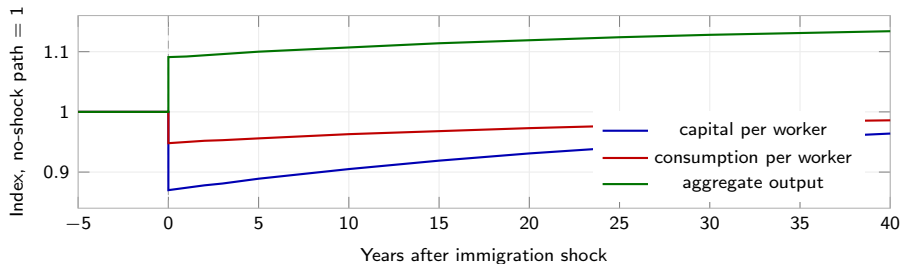
$$\text{Marginal benefit of immigration} = f(k) - kf'(k)$$

- ▶ No skill heterogeneity or selection across immigrant groups (H1b?).
- ▶ No complementarity or substitution across native and immigrant labor types.
- ▶ No endogenous assimilation, human-capital accumulation, innovation, or entrepreneurship channel.

These channels would show up through $f(\cdot)$, ξ , and transition from immigrant to native status

Transition dynamics: level shock in Solow model

Cobb-Douglas Solow economy initially at its Golden Rule steady state. At $t = 0$, the labor force rises by 15%; capital is predetermined.



- ▶ Aggregate output jumps because there are more workers.
- ▶ Capital and consumption per worker fall on impact, then recover gradually.
- ▶ 40+ years